Real-Time AI for Critical Decisions

Early deterioration detection with less alarm fatigue

Powered by IBM LinuxONE (Linux on Z)

IBM Z Datathon | Project Submission

The Problem

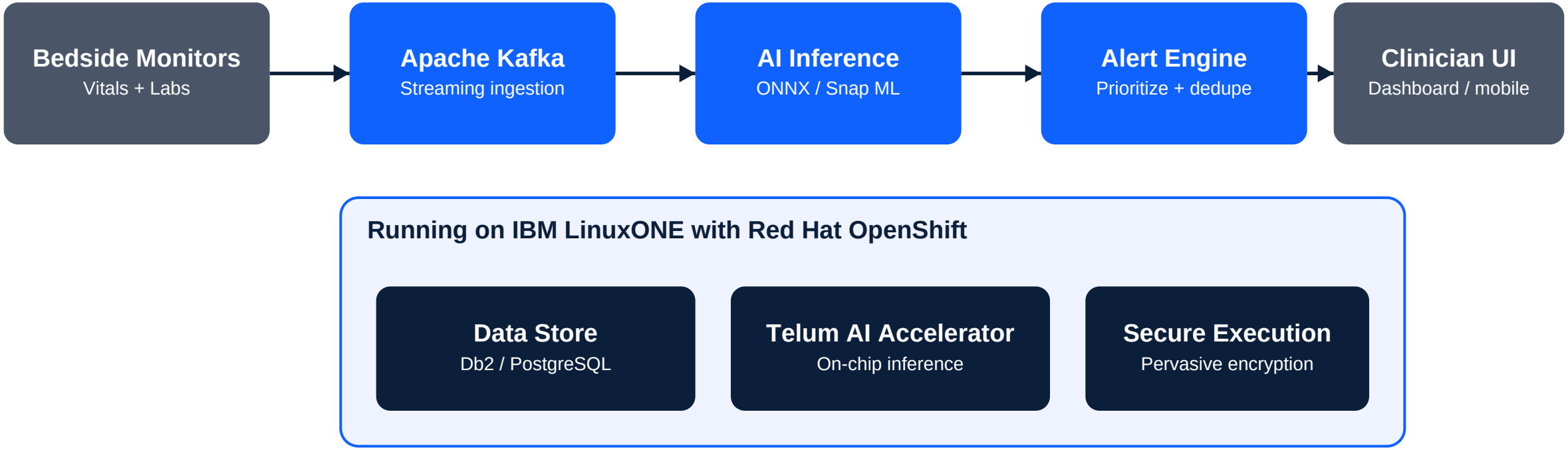
- **Late recognition kills.** Delayed detection of sepsis, cardiac arrest and respiratory failure increases mortality.
- **Alarm fatigue.** Clinicians face constant false or low-priority alarms, so real emergencies get missed.
- **Scattered signals.** Vitals, labs and history sit in separate systems with no unified, real-time risk view.
- **Hard to trust.** Healthcare AI needs low latency, high availability and strict data protection.

Our Solution

A continuous real-time risk engine that watches every patient and raises **prioritized, context-aware alerts** early.

- **Capture:** Heart rate, blood pressure, SpO2, respiratory rate, temperature and lab values stream from bedside monitors.
- **Predict:** Deterioration models score each patient continuously for sepsis, cardiac arrest and respiratory failure.
- **Prioritize:** Alerts are ranked by risk and context, filtering noise to cut false alarms.
- **Act:** Care teams get early warnings on dashboards and mobile, with time to intervene.

Solution Architecture



Models are trained offline and deployed for in-platform inference, so scoring happens next to the data with no external GPU round trips.

How We Leverage IBM Z / LinuxONE

- **Telum on-chip AI acceleration:** millisecond inference co-located with patient data.
- **Pervasive encryption and Secure Execution:** confidential computing for sensitive health data, supporting HIPAA / GDPR compliance.
- **Extreme availability:** up to 99.999% uptime so critical alerts are never missed.
- **Scale and isolation:** thousands of concurrent patient streams with consistent latency and workload isolation.
- **Open, cloud-native stack:** Kafka, OpenShift and standard ML runtimes on Linux, so no vendor lock-in for developers.

Value and Impact

Patients

Earlier intervention can reduce mortality and avoidable ICU transfers.

Clinicians

Fewer false alarms means less fatigue and more focus on patients in real need.

Hospitals

Lower costs, shorter stays, a secure, compliant platform that scales across wards and facilities.

Outcome: early, prioritized alerts give care teams more time to act when it matters most.

Roadmap and Next Steps

- **Now:** Real-time ingestion, risk scoring and prioritized alerting prototype on LinuxONE.
- **Next:** Validate on public clinical datasets and tune thresholds with clinician feedback.
- **Then:** EHR integration (HL7 / FHIR), explainable risk factors and pilot in a hospital ward.
- **Scale:** Multi-facility deployment with federated learning and continuous model monitoring.

Detect sooner. Alert smarter. Save more lives.

Real-time AI for critical decisions, built on IBM LinuxONE.

Thank you